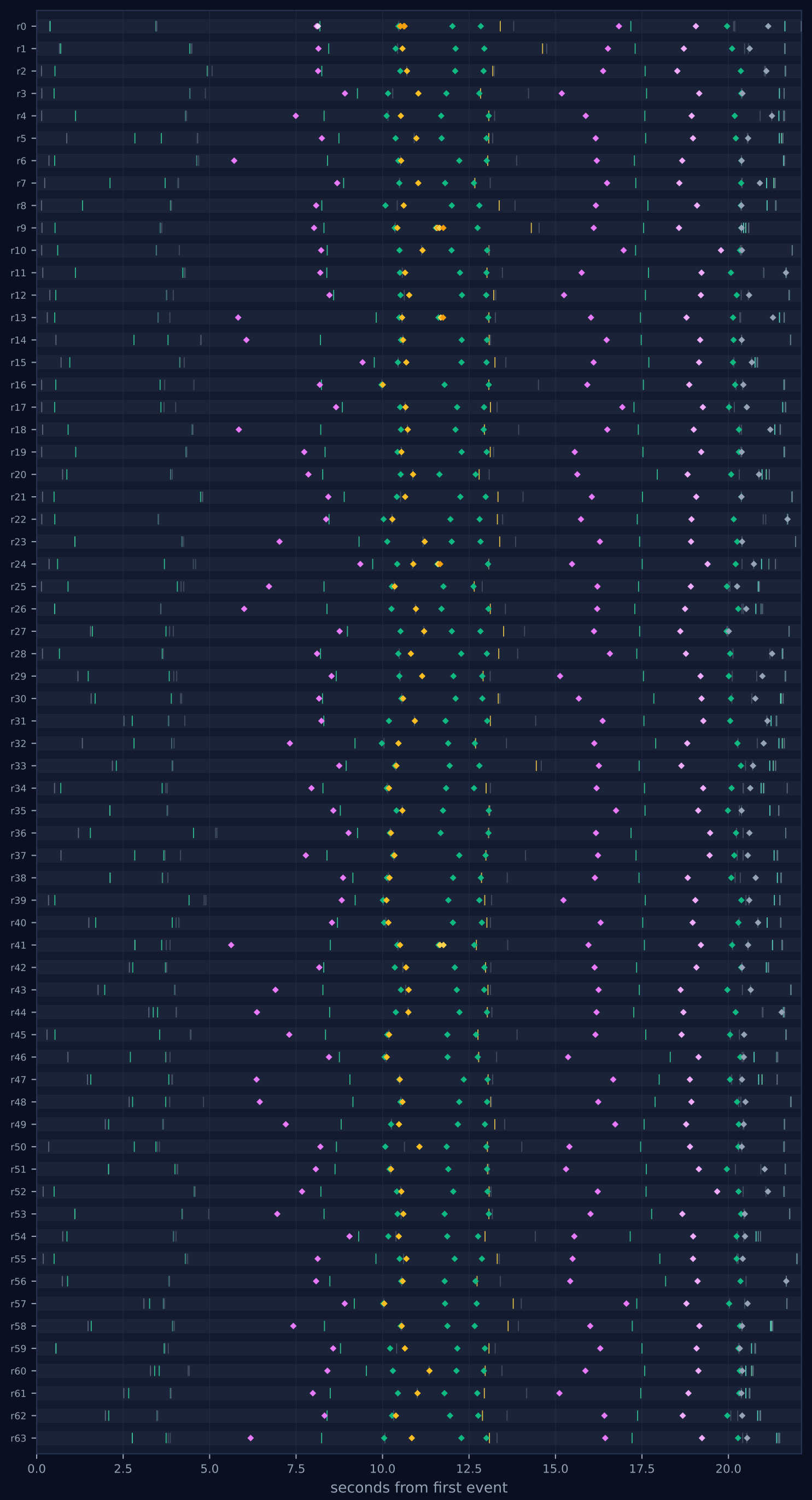
r0
r1
r2
r3
r4
r5
r6
r7
r8
r9
r10
r11
r12
r13
r14
r15
r16
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r52
r53
r54
r55
r56
r57
r58
r59
r60
r61
r62
r63
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
seconds from first event
collective
window operation
reduction / arbitration
rank lifecycle